

Review: "Thinking in the Rectified Latent Manifold: Dual-Anchored Alignment for Robust Remote Sensing Visual Grounding"

Summary

The paper proposes DAMA (Dual-Anchored Manifold Alignment), a framework for robust remote sensing visual grounding under joint image and text perturbations. The method introduces a latent reasoning interval bounded by special tokens (`<DAMA_START>`/`<DAMA_END>`), regularized by two losses: Latent Visual Alignment (LVA, aligning hidden states to ROI-derived visual embeddings via MSE) and Textual Space Alignment (TSA, a causal LM objective over the latent tokens). Training proceeds in two stages: supervised fine-tuning (DA-SFT) followed by a GRPO variant (LE-GRPO) that replays cached hidden states as fixed KV states for policy-gradient optimization over the continuous latent space. A Hybrid Multi-LoRA / mixture-of-degradation-experts (MoDE) module handles heterogeneous corruption types. Experiments are run on VRSBench and DIOR-RSVG under four image corruption types (S&P noise, blur, cloud, compression) crossed with two text-rewriting styles (Naturalistic, Persona), against GeoChat, LLaVA-V1.5, CSDNet, and Qwen2.5-VL-3B baselines.

Major Concerns

M1. Perturbation composition not reported for training or evaluation

The paper does not state whether the images used during training (DA-SFT and LE-GRPO) are perturbed or clean. On the evaluation side, the authors do not give counts or per-corruption-type sample sizes for the perturbed test sets used to compute the reported accuracies.

Suggestion: State explicitly whether training images are perturbed, and report counts or percentages per corruption type for both the training data and the evaluation sets used in Tables I-II.

M2. No evaluation on clean (unperturbed) test data

Tables I and II report accuracy only under perturbed conditions. No table reports accuracy on the original, unperturbed DIOR-RSVG or VRSBench test/validation sets, for any method.

Suggestion: Include accuracy on clean test splits for all compared methods.

M3. No comparison to a robustness-oriented method

Related Work discusses prior approaches specifically targeting multimodal robustness in RS, but none of these are included as baselines in Table I or Table II. All benchmarked methods are general-purpose grounding models fine-tuned on the same data, rather than methods designed to address the robustness problem this paper targets.

Suggestion: Include at least one robustness-focused method as a baseline, or explain why none could be included.

M4. Ablations restricted to two corruption types and one dataset

Tables III-VI (LVA/TSA, alignment projector, loss function, multi-LoRA ablations) report results only for S&P Noise and Blur, only on VRSBench. Cloud and compression conditions, and DIOR-RSVG, are absent from all ablation tables.

Suggestion: Extend the ablations to cover cloud and compression conditions, and DIOR-RSVG, or explain the restricted scope.

M5. Comparative discussion lacks analysis of result patterns

Section IV-C restates accuracy improvements without addressing two patterns visible in the tables: compression yields the lowest scores for nearly all methods in Table I, and the margin over the baselines is much smaller on DIOR-RSVG (Table II: GeoChat 31.94% vs. Ours 33.83%, Naturalistic/Avg) than on VRSBench (Table I: Qwen2.5-VL-3B 20.15% vs. Ours 36.99%, Naturalistic/Avg).

Suggestion: Discuss why compression is the hardest corruption across methods, and why the DIOR-RSVG margin is smaller than the VRSBench margin.

M6. Dataset-filtering criterion undefined and unaudited; filter model differs from benchmarked model

Section III-B states that Qwen3-VL-32B is used to "filter out invalid or unreasonable examples." No rejection rate, filtering criteria, or prompt is given. No human validation of the filter's decisions is reported. The filtering model (Qwen3-VL-32B) is also a different generation from the model used as a baseline (Qwen2.5-VL-3B, fine-tuned as a comparison method in Tables I-II).

Suggestion: Report the filtering criteria/prompt and rejection rate, include a human-validated sample of filter decisions, and clarify why a Qwen-family model was used to filter the dataset while a Qwen-family baseline is also being benchmarked.

Minor Concerns

Ambiguity in whether images are perturbed during training

Section III-B states that "standard textual instructions" are used during training, meaning text is unperturbed. Whether training images are ever perturbed is not stated.

Suggestion: Add a direct statement on whether training images are perturbed (see M1 for the proportion, if so).

Undefined "VRSBench-Ref" test split

Section IV-A names train and validation splits for VRSBench only. Section IV-C states that the models are evaluated on "VRSBench-Ref," a term used once and not otherwise defined.

Suggestion: Define VRSBench-Ref and clarify its relation to the splits in IV-A.

Metric threshold not justified

Only IoU@0.4 is reported throughout. No justification is given for this threshold, and no results at other thresholds (e.g., 0.5, 0.7) or mean IoU are shown.

Suggestion: Justify the choice of 0.4, or add results at more standard thresholds.

Typographical errors

- "acheive"
- "pertrubations," "distructaion," "envirement"
- "mainfold" (inconsistent with "manifold" used elsewhere)
- "performace"
- "approachs emphasis"
- Fig. 6 caption, last row: transcribed query text ends with "Assistant:"

Questions for the Authors

1. What proportion of DA-SFT and LE-GRPO training samples use perturbed images, and which corruption types are represented in training?
2. What is "VRSBench-Ref," and how does it relate to the splits described in Sec. IV-A?
3. Can results on clean test data be added for all compared methods?
4. Why was no robustness-focused method included as a baseline?
5. Why does compression yield the lowest accuracy across nearly all methods (Table I)?
6. Why is the margin over the baselines smaller on DIOR-RSVG (Table II) than on VRSBench (Table I)?

7. What are the filtering criteria used by Qwen3-VL-32B, what fraction of perturbations were rejected, and was any human validation performed? Could using a Qwen-family model for filtering bias the dataset relative to the Qwen2.5-VL-3B baseline?
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Recommendation

Major revision. The core idea is reasonably well-motivated, but the paper would benefit from clarifying the training data composition (M1), adding clean-condition results (M2), comparing against a robustness-focused baseline (M3), extending the ablations (M4), discussing the comparative results in more depth (M5), and documenting/validating the dataset filtering process (M6).